Building a World-Class ML Model for Dyslexic & ADHD Students in Literature

The Core Problem

A student with dyslexia opens *Things Fall Apart* by Chinua Achebe. They hit: *"He was tall and huge, and his bushy eyebrows and wide nose gave him a very severe look."* They freeze on "eyebrows," the letters swim, they lose the sentence, they lose the paragraph, they lose interest. A student with ADHD reads the same page three times and retains nothing because their attention drifted at line 2.

The model's job: keep that student in the story.

Deliverable 1: Real-Time Struggle Detection Engine

Forget quizzes after the fact. The model must detect struggle **as it happens**.

Signals to capture:

Signal	What it really means
Eye-tracking / gaze proxy (camera)	Which exact word caused a fixation or regression
Reading velocity changes	Sudden slowdown = decoding difficulty
Touch pressure (mobile)	Frustration correlates with harder taps
Pause duration between page turns	Processing overload vs. genuine engagement
Re-reading patterns	Scrolling back = lost comprehension
Time-of-day + session length	ADHD attention degrades predictably
Highlight/tap behavior	Student self-identifies difficult words

The ideal model: A **multimodal transformer** that fuses all these signals into a single "struggle score" updated every 2–3 seconds — not every page, not every chapter. Real-time.

Deliverable 2: Instant "What Does This Mean?" — Context-Aware Simplification

This is your highlight feature, but done properly:

Student highlights: *"The drums were still beating, persistent and unchanging. Their sound was no longer a separate thing from the living village."*

Bad simplification: "The drums kept playing. The sound was part of the village."
(Kills the literature. Kills the learning.)

World-class simplification:

> ***Simpler version:*** *"The drums kept beating steadily. After a while, their sound felt like it belonged to the village itself — like it was alive, just like the people."*
> ***Why Achebe wrote it this way:*** *He's showing that in Igbo culture, music isn't entertainment — it's part of life itself. The drums are the village.*
> ***Key word — "persistent":*** *means something that keeps going and won't stop. Think of a persistent alarm clock.*

What the model must do:

- Simplify **without destroying literary voice**
- Explain the **author's intent** (why it matters to the story)
- Give **vocabulary support** with relatable analogies

- Adjust the explanation level based on the student's **reading profile**
- Offer a **Kinyarwanda bridge** — connect the concept to something culturally familiar

Architecture: Fine-tuned LLM (distilled for edge deployment) with:

- Novel-specific context window (knows the full book)
- Student reading level embedding
- Literary device recognition (metaphor, irony, foreshadowing)

Deliverable 3: Adaptive Reading Mode Engine

Not one adaptation — a **dynamic combination** that shifts in real-time:

For Dyslexia:

Adaptation	When to trigger
Bionic Reading (bold first syllables)	Decoding slowdown detected
OpenDyslexic font switch	Persistent letter-reversal patterns
Syllable color-coding	Long word fixation (>3 seconds on a word)
Line spacing increase	Vertical tracking errors
Reading ruler (highlight current line)	Line-skipping detected
Audio narration of current sentence	Complete decoding breakdown

For ADHD:

Adaptation	When to trigger
Chapter chunking (break into 3-min segments)	Session start — always
Progress gamification ("5 paragraphs to next checkpoint!")	Attention drift detected
Inline comprehension micro-checks	Every 2–3 paragraphs
Ambient focus sounds	Student-selected preference
"Story so far" recap	Returning after a break
Cliffhanger breaks ("Want to know what Okonkwo does next?")	Natural story tension points

The RL policy network decides the combination and intensity. Not rules. Not if-then. A learned policy that knows *this particular student* at *this particular moment* needs *this particular cocktail* of adaptations.

Deliverable 4: Literature Comprehension Graph

A knowledge graph built per student, per book:

```
Student reads "Things Fall Apart"
    │
    ├── Characters understood: Okonkwo ✓, Ikemefuna ✓, Ezinma ~partial
    ├── Themes tracked: Masculinity ✓, Colonialism ✗ (not yet encountered)
    ├── Literary devices recognized: Proverbs ✓, Irony ✗
```

```
    ├── Vocabulary acquired: 47/120 target words
    ├── Chapters with >70% comprehension: 1,2,3,5
    ├── Chapters needing revisit: 4 (comprehension: 45%)
    └── Predicted struggle zones: Chapters 13–15 (vocabulary spike +
         narrative complexity increase)
```

Why this matters: The model doesn't just react — it **anticipates**. Before the student reaches Chapter 13, it pre-teaches vocabulary. It surfaces a "Previously in the story..." recap for complex plot threads.

Architecture: Graph neural network updated after every reading session, connected to a book-level difficulty map generated at content upload time.

Deliverable 5: The Student Reading Profile (Learner Embedding)

A **128-dimensional vector** that encodes everything about how this student reads:

- Decoding speed by word length
- Attention span curve (minutes before drift)
- Preferred learning modality (visual / auditory / read-write)
- Vocabulary level (per subject domain)
- Emotional response to difficulty (gives up fast vs. persists)
- Time-of-day performance patterns
- Which adaptations historically helped vs. annoyed them

This vector is the **input to every model decision**. It's what makes the system personalized, not generic. It updates every session. It transfers across books.

Deliverable 6: Teacher Intelligence Layer

Teachers don't want dashboards with graphs. They want answers:

> *"**Amina** is 2 chapters behind the class. She understands characters well but struggles with figurative language. **Recommended action:** Pair her with Jean for the proverbs discussion on Thursday."*

> *"**3 students** highlighted the same passage in Chapter 7. They may not understand the concept of tragic irony. **Suggested:** 5-minute class discussion before moving to Chapter 8."*

The model generates:

- Natural language student summaries (not charts)
- Specific, actionable teaching recommendations
- Class-wide pattern alerts (if many students struggle on the same thing)
- Predicted exam risk areas

Deliverable 7: Offline-First Inference Stack

Rwanda reality: intermittent power, shared devices, limited bandwidth.

Component	Spec
Core RL policy	< 20MB, runs on $50 Android phone
Simplification model	Distilled 1B-parameter LLM, < 500MB, quantized to INT4
Book content + difficulty maps	Pre-computed, cached per novel (~5MB/book)

Component	Spec
Sync strategy	Delta-sync when online — only new session data uploads
Cold start	Works immediately with population-level defaults, personalizes within 3 sessions

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The Ideal Architecture (Summary)

```
┌──────────────────────────────────────────────────────┐
│                   STUDENT DEVICE                     │
│                                                      │
│  ┌──────────┐    ┌──────────────┐    ┌──────────┐   │
│  │ Struggle │──▶│  RL Policy   │──▶│Adaptation│   │
│  │ Detector │    │ (PPO Agent)  │    │ Renderer │   │
│  └──────────┘    └──────────────┘    └──────────┘   │
│       ▲                │                             │
│       │            ┌───▼────────┐                    │
│  Raw signals       │  Student   │                    │
│  (scroll, tap,     │ Embedding  │                    │
│   pause, gaze)     │ (128-dim)  │                    │
│                    └────────────┘                    │
│                                                      │
│  ┌──────────────────────────────────────────┐        │
│  │  On-Device Simplification LLM (distilled)│        │
│  │  + Book Context Window + Literary Device  │        │
│  │    Recognition + Kinyarwanda Bridge       │        │
│  └──────────────────────────────────────────┘        │
└──────────────────────────────────────────────────────┘
                    │ (when online)
              ┌─────▼──────┐
              │ Cloud Sync  │
              │ + Retrain   │
              │ + Teacher   │
              │ Dashboard   │
              └────────────┘
```

What Makes This World-Class

1. **It preserves the literature.** Simplification doesn't dumb down — it bridges understanding while teaching the original language.
2. **It's invisible when working.** The best adaptation is the one the student doesn't notice. No pop-ups. No "you seem to be struggling!" Just quietly adjusting.
3. **It learns the individual, not the diagnosis.** Not all dyslexic students are the same. The embedding captures *this* student.
4. **It anticipates, not just reacts.** The comprehension graph predicts struggle before it happens.
5. **It works without internet.** Because Rwanda. Because reality.
6. **It gives teachers superpowers**, not homework. Actionable insights in plain language.

The highlight-to-simplify feature you mentioned is Deliverable 2 — and it's the most immediately impactful one to build first. Want me to start implementing it in the codebase?

So another problem with the system it might face when a teacher uploads a pdf of the learning content, it is not displayed in a User friendly way especially for these specific groups of students. So how do i make sure plays, novels and any other literature resource no matter how complex is properly analyzed by the Machine Learning model and given back to the student in a way they can not only understand but enjoy.

The PDF-to-Accessible-Reading Pipeline

The core challenge: a teacher uploads a scanned PDF of *Romeo and Juliet* or a Rwandan novel, and what the student sees must feel like it was **designed for them** — not like a PDF viewer.

The Problem Visualized

```
WHAT THE TEACHER UPLOADS          WHAT THE STUDENT CURRENTLY SEES

┌──────────────────────┐          ┌──────────────────────┐
│ Romeo and Juliet     │          │ Romeo and Juliet     │
│ Act 1, Scene 1       │          │ Act 1, Scene 1       │
│                      │          │                      │
│ SAMPSON: Gregory,    │    ──►   │ SAMPSON: Gregory,    │
│ o' my word, we'll    │          │ o' my word, we'll    │
│ not carry coals.     │          │ not carry coals.     │
│                      │          │ [Tiny text, no       │
│ GREGORY: No, for     │          │  structure, just a   │
│ then we should be    │          │  wall of words]      │
│ colliers.            │          │                      │
└──────────────────────┘          └──────────────────────┘

        ↓ WHAT IT SHOULD BECOME ↓

┌──────────────────────────────────────────────────┐
│  Romeo & Juliet  ·  Act 1  ·  Scene 1            │
│  ═══════════════════════════════════════════     │
│  📍 A street in Verona. Two servants enter.      │
│                                                  │
│  ┌──────────┐                                    │
│  │ SAMPSON  │  "Gregory, o' my word, we'll       │
│  │ 🗡 Capulet│  not carry coals."                 │
│  └──────────┘                                    │
│       💬  ┌──────────────────────────┐           │
│           │ What he means: "We won't  │           │
│           │ take insults from anyone." │           │
│           │                          │           │
│           │ "carry coals" = old English │        │
│           │ for accepting humiliation  │           │
│           └──────────────────────────┘           │
│                                                  │
│  ┌──────────┐                                    │
│  │ GREGORY  │  "No, for then we should be        │
│  │ 🗡 Capulet│  colliers."                        │
│  └──────────┘                                    │
│       💬 He's making a joke — "colliers"         │
│          means coal workers. He's playing        │
```

```
│          on Sampson's words.                    │
│                                                 │
│  ▶ 🔊  Listen to this scene                     │
│  ▓▓▓
```

```
  "structure": [
    {
      "type": "act",
      "number": 1,
      "scenes": [
        {
          "type": "scene",
          "number": 1,
          "setting": "A public place in Verona",
          "characters_present": ["Sampson", "Gregory"],
          "estimated_read_time_minutes": 8,
          "chunks": [
            {
              "type": "dialogue",
              "speaker": "Sampson",
              "faction": "Capulet",
              "text": "Gregory, o' my word, we'll not carry coals.",
              "difficulty_score": 0.82,
              "archaic_phrases": [
                {
                  "phrase": "carry coals",
                  "modern_meaning": "accept insults or humiliation",
                  "context_note": "This was a common insult in Elizabethan England"
                }
              ],
              "literary_devices": ["wordplay_setup"],
              "emotion": "boastful"
            }
          ]
        }
      ]
    }
  ]
}
```

Stage 2: Pre-Computation — The Book Brain

Before the student reads a single word, the model pre-analyzes the **entire book**:

A. Difficulty Mapping

```
Chapter/Scene Difficulty Curve for "Things Fall Apart"

Difficulty
1.0 |          ┌─┐              ┌───┐
    |         /   \            /     \  ← Ch.13-15
0.8 |   ┌─┐  /     \     ┌─┐/        \    (Okonkwo's exile
    |  /   \/       \   /   \          \   + missionaries arrive)
0.6 | /               \/     \          \
    |/                \       \          \
0.4 |                          \         \
    |                                     \
0.2 |
```

```
     └──────────────────────────────────────────────
      Ch.1  3   5   7   9   11  13  15  17  19
```

For every chunk, pre-compute:

- **Vocabulary difficulty** — word frequency analysis against student's grade level
- **Sentence complexity** — clause depth, passive voice density
- **Conceptual difficulty** — abstract themes, cultural references
- **Predicted struggle points** — specific words/phrases likely to cause problems

B. Character & Relationship Map

Auto-generated and updated as the student progresses:

```
┌──────────┐   father of    ┌──────────┐
│ OKONKWO  │──────────────→│  NWOYE   │
│ (warrior)│               │  (son)   │
└──────────┘               └──────────┘
     │ kills                    │
     ▼                          │ converts to
┌──────────┐               ┌──────────┐
│ IKEMEFUNA│               │CHRISTIANITY│
│ (foster  │               └──────────┘
│  son)    │
└──────────┘
```

The student can **tap any character name** in the text and see: who they are, their relationships, and a "story so far" for that character. No more "wait, who is this person again?"

C. Cultural Context Bank

For Rwandan students reading Shakespeare, or any cross-cultural text:

> ***"carry coals"*** *— In Shakespeare's time, coal carriers were looked down on. Sampson is saying "we won't be treated like lowly workers." Think of it like someone saying "we won't be disrespected" in Kigali today.*

Pre-generated for every culturally distant reference in the book.

Stage 3: The Adaptive Rendering Engine

This is where the parsed structure becomes a **living reading experience**:

A. Genre-Specific Display Templates

```
┌──────────────────────────────────────────┐
│              NOVEL MODE                  │
│                                          │
│  Clean paragraph flow                    │
│  Character avatars in dialogue           │
│  Chapter progress bar                    │
│  "Story so far" recap at chapter start   │
│  Vocabulary sidebar (toggleable)         │
└──────────────────────────────────────────┘


┌──────────────────────────────────────────┐
│              PLAY MODE                   │
│                                          │
```

```
│  Script-style layout                      │
│  Character cards with faction colors      │
│  Stage directions in italic boxes         │
│  Scene visualizer (who is on stage?)      │
│  "What just happened" after each scene    │
└──────────────────────────────────────────┘


┌──────────────────────────────────────────┐
│              POEM MODE                    │
│                                          │
│  Centered verse with breathing room       │
│  Line-by-line audio playback              │
│  Rhyme scheme highlighting                │
│  Meaning layer (tap to reveal)            │
│  Emotional tone color gradient            │
└──────────────────────────────────────────┘
```

B. The Dyslexia Rendering Layer

Every text chunk passes through:

```
Raw text chunk
      │
      ▼
┌──────────────────────────────────────┐
│  DYSLEXIA RENDERING PIPELINE         │
│                                      │
│  1. Font → OpenDyslexic / Lexie      │
│  2. Line spacing → 1.8x minimum      │
│  3. Letter spacing → +2px            │
│  4. Line length → max 60 chars       │
│  5. Paragraph chunking → 3-4 lines   │
│  6. Syllable markers on hard words   │
│  7. Alternating line backgrounds     │
│     (subtle gray/white bands)        │
│  8. Reading ruler follows scroll     │
└──────────────────────────────────────┘
```

C. The ADHD Engagement Layer

```
Full chapter (20 pages)
      │
      ▼
┌──────────────────────────────────────┐
│  ADHD CHUNKING ENGINE                │
│                                      │
│  Break into 2-3 minute segments      │
│  Each segment ends with:             │
│    → Micro-quiz (1 question)         │
│    → Progress reward animation       │
│    → "What happens next" teaser      │
│                                      │
│  Between segments:                   │
```

```
│    → 10-second breathing break      │
│    → Choice: continue or bookmark   │
│                                     │
│  Gamification:                      │
│    → Reading streak counter          │
│    → Chapter completion badges       │
│    → "You understood 8/10 scenes!"  │
└─────────────────────────────────────┘
```

Stage 4: The Live Highlight-to-Understand Feature

The flagship interaction — student highlights anything, gets instant help:

```
Student highlights: "we'll not carry coals"
                            │
                            ▼
              ┌──────────────────────────┐
              │   CONTEXT ANALYZER       │
              │                          │
              │ What was highlighted?    │
              │ → Idiomatic expression   │
              │                          │
              │ Where in the book?       │
              │ → Act 1, Scene 1, line 1│
              │ → Sampson speaking       │
              │ → Pre-fight tension      │
              │                          │
              │ Student's level?         │
              │ → Reading level: Grade 4 │
              │ → Vocabulary: developing │
              │ → Prefers: examples      │
              └──────────────┬───────────┘
                             ▼
              ┌──────────────────────────┐
              │   SIMPLIFICATION LLM     │
              │   (book-aware, student-  │
              │    aware)                │
              └──────────────┬───────────┘
                             ▼
     ┌──────────────────────────────────────┐
     │                                      │
     │  "We won't take insults"             │
     │                                      │
     │  ────────────────────────────        │
     │  "Carry coals" is an old English     │
     │  expression. Coal carriers were       │
     │  seen as low-status workers.         │
     │                                      │
     │  Sampson is saying: "Nobody messes   │
     │  with us." He's trying to sound      │
     │  tough in front of Gregory.          │
     │                                      │
```

```
    │  🔊 Hear it    📖 See it used again    │
    │  ✏️ Add to my vocabulary list          │
    └──────────────────────────────────────┘
```

Key design rules:

1. **First line: the simple meaning** — always. No student should have to read a paragraph to get the point.
2. **Second block: the why** — context, author intent, cultural background. Optional to read.
3. **Action buttons** — hear it, save it, see other examples. Student controls depth.
4. **Never break immersion** — the explanation appears as an overlay, not a new page. One tap to dismiss and keep reading.

Stage 5: The Feedback Loop

Every interaction feeds back into the model:

```
Student highlights "carry coals"
        │
        ├──▶ Log: vocabulary_gap detected
        │    (archaic_idiom, Elizabethan English)
        │
        ├──▶ Update student embedding:
        │    idiom_comprehension -= 0.1
        │
        ├──▶ RL Agent receives signal:
        │    "Pre-teach idioms before next scene"
        │
        └──▶ Teacher dashboard:
             "3 students struggled with Elizabethan
              idioms in Act 1. Consider a 5-min
              vocabulary warm-up before Act 2."
```

The Complete Pipeline

```
  TEACHER UPLOADS PDF
        │
        ▼
  ┌──────────────┐
  │  OCR + Parse  │ ← Extracts raw text from any PDF
  └──────────────┘
        │
        ▼
  ┌──────────────┐
  │  Literary     │ ← Identifies: novel/play/poem,
  │  Structure    │   chapters, dialogue, characters,
  │  Recognition  │   stage directions, verse/prose
  └──────────────┘
        │
        ▼
  ┌──────────────┐
  │  Book Brain   │ ← Pre-computes: difficulty map,
  │  Pre-Analysis │   character graph, cultural notes,
  │               │   vocabulary list, predicted struggles
  └──────────────┘
        │
        ▼
```

```
┌──────────────┐
│  Structured  │ ← Clean JSON: every sentence tagged
│  Content DB  │   with metadata, difficulty, context
└──────────────┘
       │
       ▼

┌──────────────────────────────────────┐
│        STUDENT READING SESSION       │
│                                      │
│  Genre template (novel/play/poem)    │
│  + Dyslexia rendering layer          │
│  + ADHD engagement layer             │
│  + Live highlight-to-understand      │
│  + RL agent adjusting in real-time   │
│  + Comprehension tracking            │
└──────────────────────────────────────┘
       │
       ▼

┌──────────────┐
│  Teacher gets │ ← "Here's what your class
│  insights    │    struggled with today"
└──────────────┘
```

What to Build First

Priority	Component	Why first
1	PDF parser + literary structure recognition	Nothing works without clean structured content
2	Genre-specific display templates	Students must see clean, accessible text immediately
3	Highlight-to-understand (simplification LLM)	Highest impact single feature — students self-serve
4	Dyslexia rendering layer	Font, spacing, syllable markers — measurable impact
5	ADHD chunking engine	Breaks content into digestible pieces
6	Book Brain pre-analysis	Difficulty maps, character graphs, cultural context
7	RL policy for real-time adaptation	Ties everything together intelligently

Want me to start implementing the PDF parsing and literary structure recognition pipeline in the codebase? That's the foundation everything else depends on.